



NOZOMI SIMULATOR 2.0

-

USER MANUAL

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Change history / Versions

Version	Modified	by	Date
1.00	First version	CRM	12/03/2019
2.00	Review and update	IGA	01/04/2019
2.01	Encoder configuration	IGA	22/06/2019

1. First Steps

1.1. Connect

- Local connection.

We need to configure our laptop IP in range, 192.168.8. xx. (Ex: 192.168.8.115)

- Wi-Fi connection.

We need to access at Wi-Fi emitted by the cabinet with our wireless IP in automatic mode.

WIFI: Simulator2.0_Cx

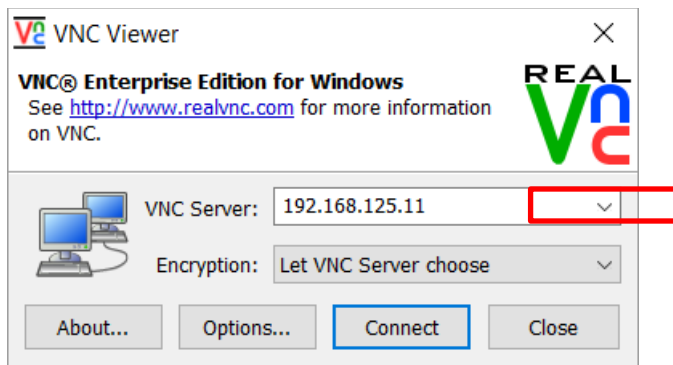
Pass: Simulator2.0

1.2. VNC Viewer

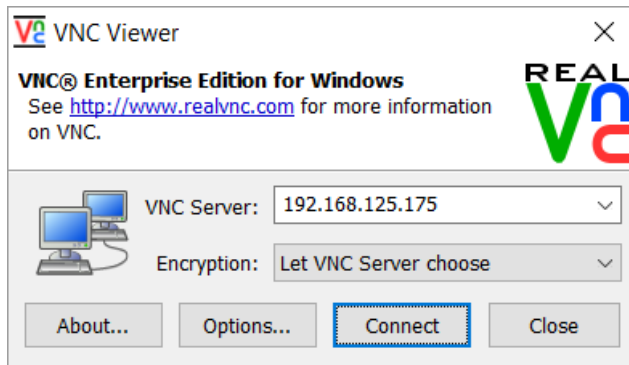
The customer can access the PLC visualization from **VNC® Viewer**.

Virtual Network Computing (VNC) is a graphical desktop-sharing system used to remotely control another computer. It transmits the keyboard and mouse events from one computer to another, relaying the graphical-screen updates back in the other direction, over a network.

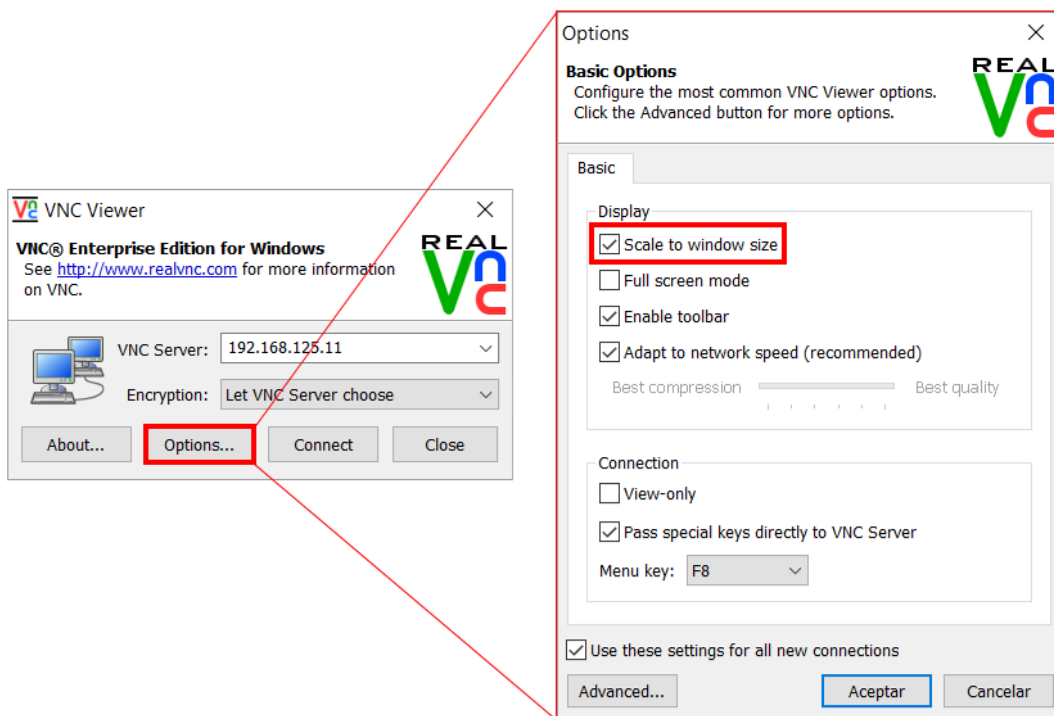
Once VNC® Viewer has been installed and executed, it is necessary to enter the IP address of the PLC. By default, the first time the PLC is switched on, its IP address is set to **192.168.125.11**.



However, in some cases we need to change the IP address of the PLC in order to establish a Modbus communication with the machine we want to test. If we make this change when connected through VNC, we will lose the connection and will need to restart it with the new IP address.



The visualization has been designed for a display size of 800x600 pixels, but it can be scaled to the window size by checking the “**Scale to window size**” checkbox in the VNC® Viewer Options.

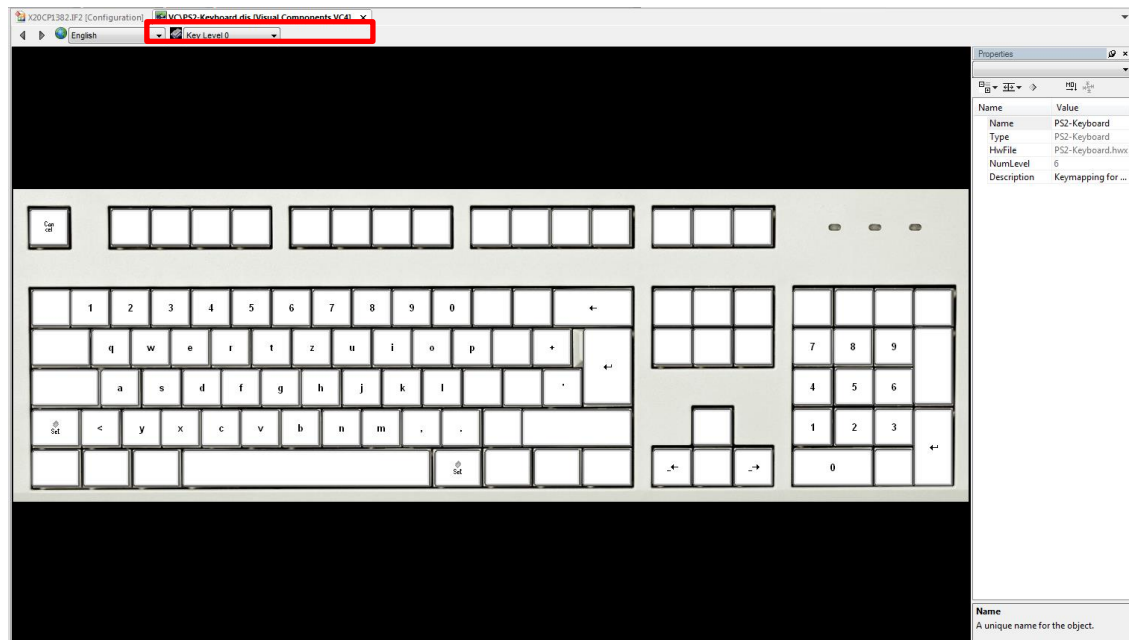
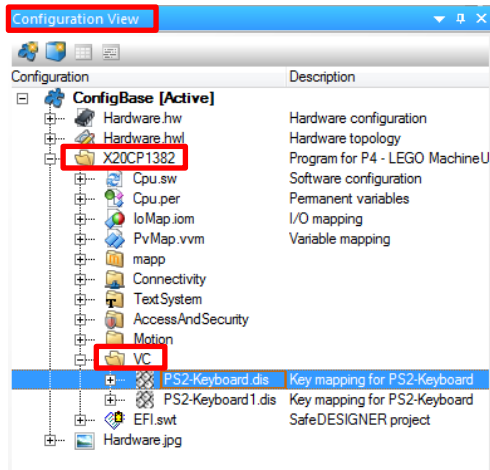


Finally, once we have entered the IP address of the PLC and enabled the scale option, we can access the PLC visualization.

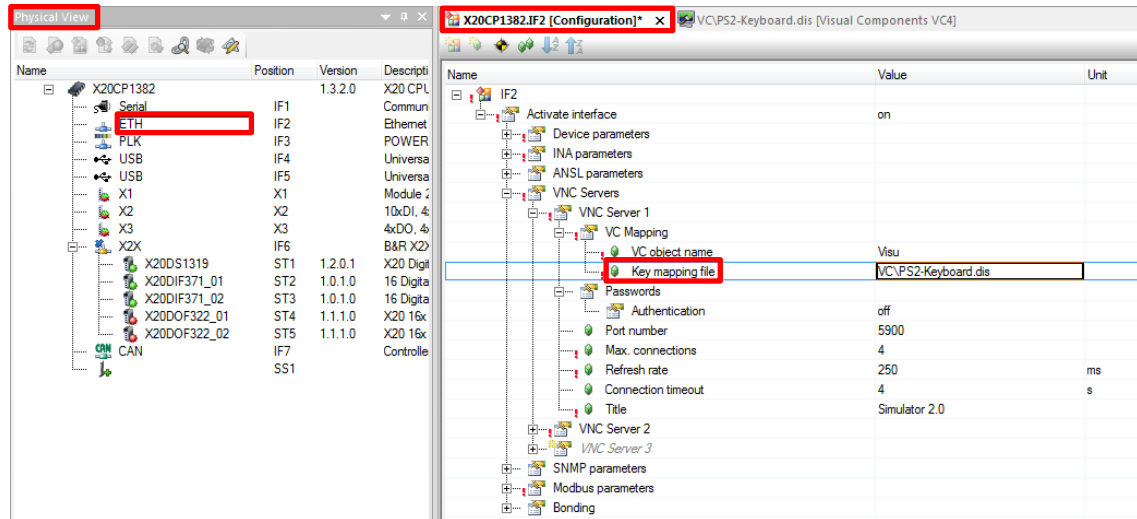
1.3. Keyboard Configuration

So that the PLC can interpret the keyboard events transmitted by the VNC from the computer, it is necessary to open the project and map the keyboard from B&R Automation Studio.

The keyboard is mapped from the configuration view of the project explorer, following the path CPU > VC, and double-clicking on 'PS2-Keyboard.dis'



Once the keyboard has been mapped according to our needs, it is necessary to select 'PS2-Keyboard.dis' as the Key mapping file. We do this from the physical view of the project explorer, by right clicking on ETH and following the path Configuration > VNC Server 1 > VC Mapping.



2. General Concepts

2.1. User Roles

There are, basically, two ways to operate the simulator: **Printer mode** and **Ancillary mode**. In the former, the PLC simulates the printer in order to test the correct operation of the ancillary machines, that is, feeder, coater and stacker. In the latter, the PLC simulates the three ancillary machines in order to test the correct operation of a printer.

With the intention of limiting the access of users to a single mode depending on whether they are providers of ancillary machines or printer producers, different user roles have been created. The following table shows the permissions of each role:

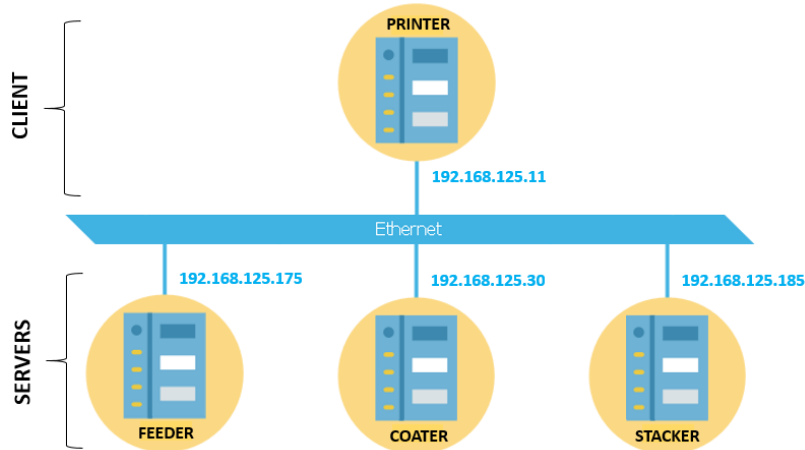
Role	Mode of operation		Machine to test			
	Printer	Ancillary	Printer	Feeder	Coater	Stacker
Administrator	X	X	X	X	X	X
Producer		X	X			
Provider	X			X	X	X
Feeder Provider	X			X		

<i>Coater Provider</i>	X				X	
<i>Stacker Provider</i>	X					X

In addition, users with administrator role have administrator permissions, which means that they have access to the user manager and the date and time settings. The former manages the creation of new users, as well as the editing, deletion and blocking of existing users. The latter adjusts the time and date of the PLC.

2.2. Modbus TCP/IP Communication

The printer communicates with the ancillary machines through a Modbus TCP/IP communication. Modbus TCP/IP uses Transmission Control Protocol and Internet Protocol for the transmission of messages from Modbus between compatible devices over various systems. That is, Modbus TCP/IP uses a physical network (Ethernet), with a networking standard (TCP/IP), and itself offers a method of representing data (Modbus as the application protocol).



As the picture above shows, the printer is the client and the ancillary machines are the servers. That means the printer is the master and requests information from the ancillary machines. In order to do that, the printer needs to know the feeder, coater and stacker IP addresses.

Since, depending on the operating mode, the PLC sometimes simulates the printer and other times the ancillary machines, it will also have to switch its role in Modbus TCP/IP communication, sometimes it will be client and others it will be server.

2.2.1. Modbus Communication in Printer Mode

In this mode, the PLC simulates the printer, so it will be the client in the modbus communication. In order to be able to communicate with the ancillary machines, the PLC needs to know their IP addresses. It does not matter what IP address the PLC has, as long as it does not coincide with any of the IP addresses of the ancillary machines.

2.2.2. Modbus Communication in Ancillary Mode

In this mode, the PLC simulates the ancillary machines, so it will be the server in the modbus communication. Now, the PLC does not need to know the client IP address, but in order to allow the client to communicate with it, the PLC needs to have one of the IP address that the printer expects for the ancillary machines.

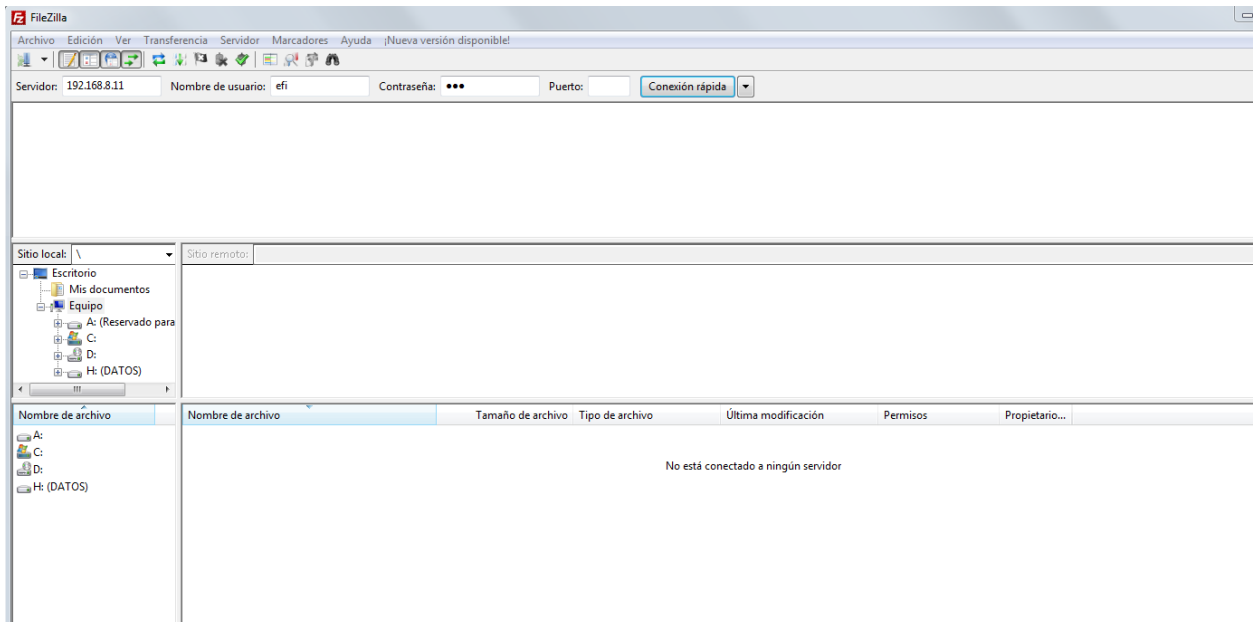
However, Modbus communication cannot be enabled for all ancillary machines simultaneously because, in order to do that, we would need the PLC to have three different IP addresses at the same time. Therefore, as regards modbus communication, in ancillary mode the modbus communication must be tested separately.

2.3. Csv file access

We can access with a FTP client (Ex. Filezilla), with the following credentials.

User: efi

Pass: efi



The file name is format is:

PrinterMode_%Y_%m_%d_%H_%M_%S.csv or AnciMode_%Y_%m_%d_%H_%M_%S.csv
depending with simulator mode we have configured.

3. Visualization.

3.1. Starting Screen

User access, date settings and user administration are managed from the start screen. For security reasons, date settings and user administration are only accessible by users with administrator permissions.

3.1.1. Login

On this screen, users can log in by typing the user name and the password in the corresponding fields. The following section shows how to create these users, as well as edit the existing ones.

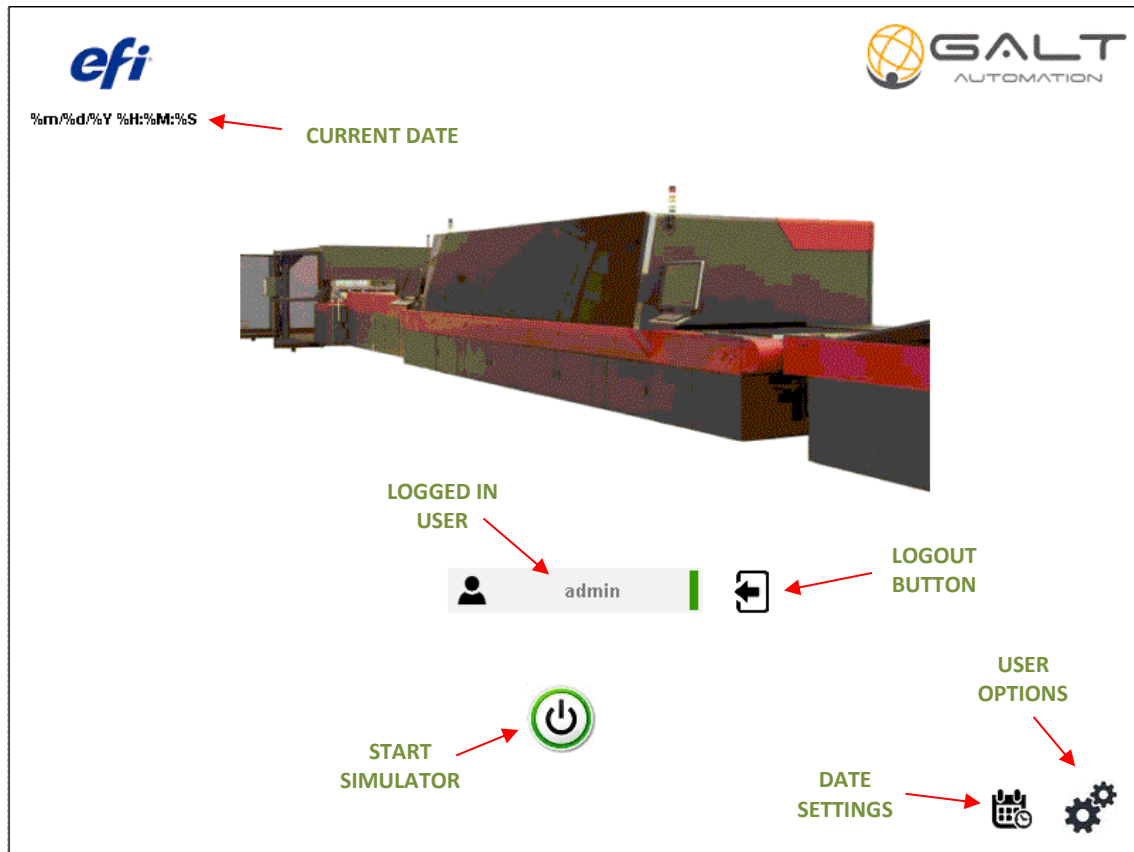


Once these two fields are filled in, we can log in by clicking on the login button. As long as the user exists and its password is valid, the login will be successful and the appearance of the screen will change, as shown below.

From this screen, we can perform up to five actions, depending on the permissions of the logged in user.

1. Log out by clicking on the logout button.
2. Start the simulator by clicking on the start simulator button.
3. Change the password of the current user from the user options screen.
4. Open the user manager, also from the user options screen.
5. Set the time and date from the date settings screen.

The first three options do not have access restrictions, but the last two (user manager and date settings) are only accessible by users with administrator permissions.



3.1.2. User Options

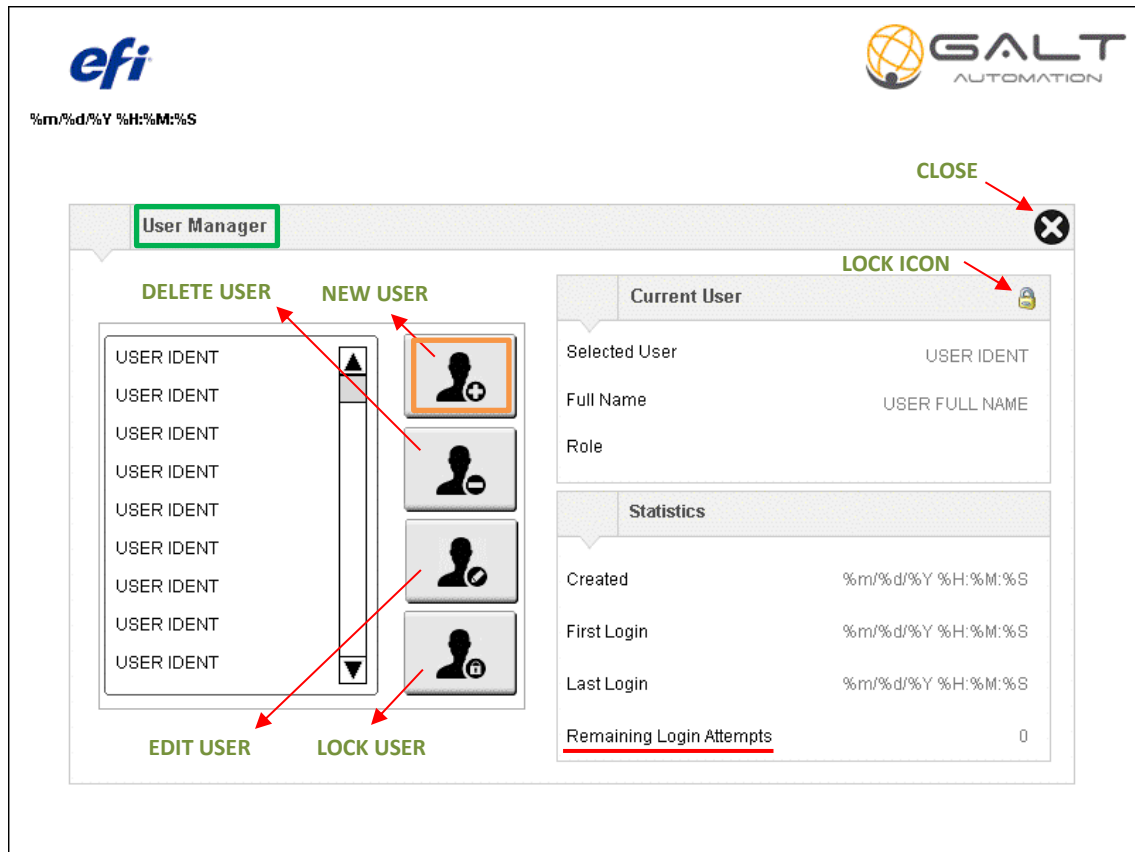
From this screen, the logged in user can change their current password by clicking on the *Change Password* button. If they have administrator permissions, the *User Manager* button will be enabled so they will be able to access the user manager by clicking on it.



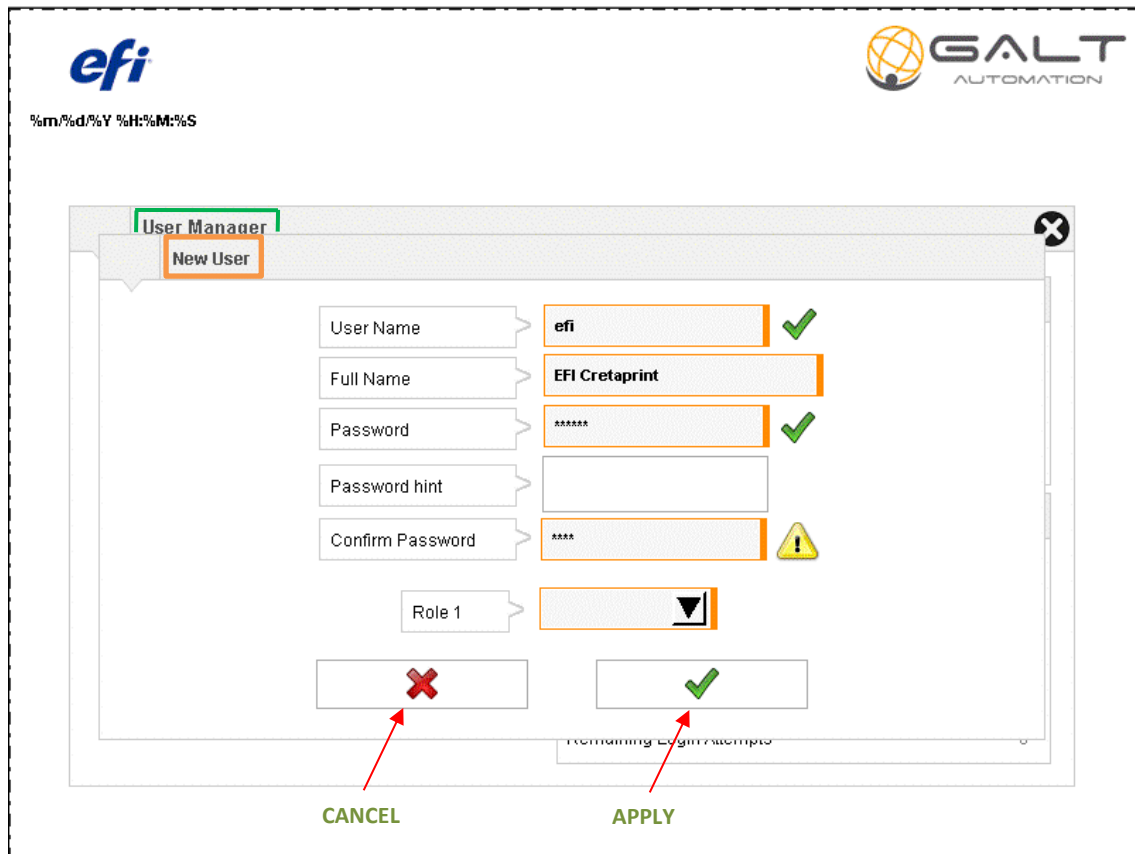


The User Manager is only available for users with administrator permissions. From this screen, these users can:

- Create new users.
- Delete existing users.
- Edit existing users.
- Lock existing users



The data shown in the *Current User* and *Statistics* boxes refers to the user selected on the left-hand list. As is shown in the following image, each button displays a new window from which the corresponding action can be performed by filling in the appropriate fields.



efi

%m/%d/%Y %H:%M:%S

GALT
AUTOMATION

User Manager

New User

User Name: efi ✓

Full Name: EFI Cretaprint

Password: ***** ✓

Password hint:

Confirm Password: **** !

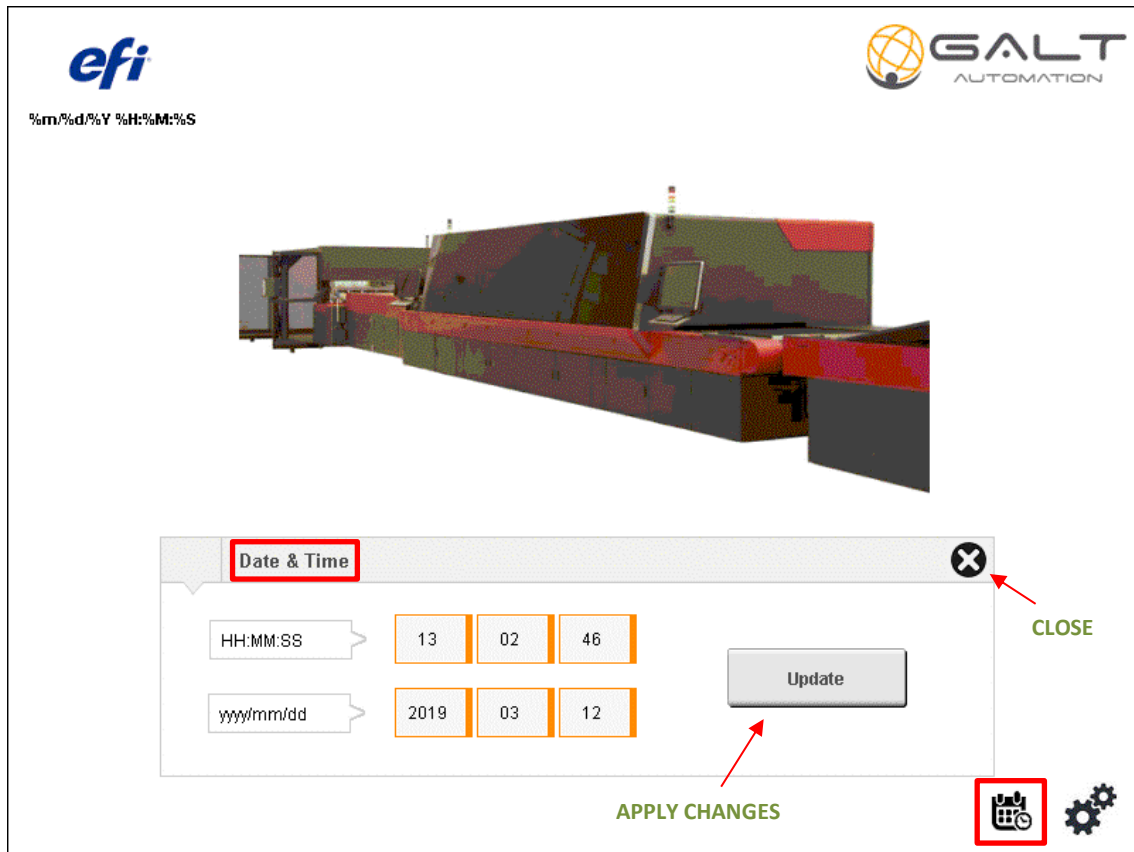
Role 1: ▼

CANCEL

APPLY

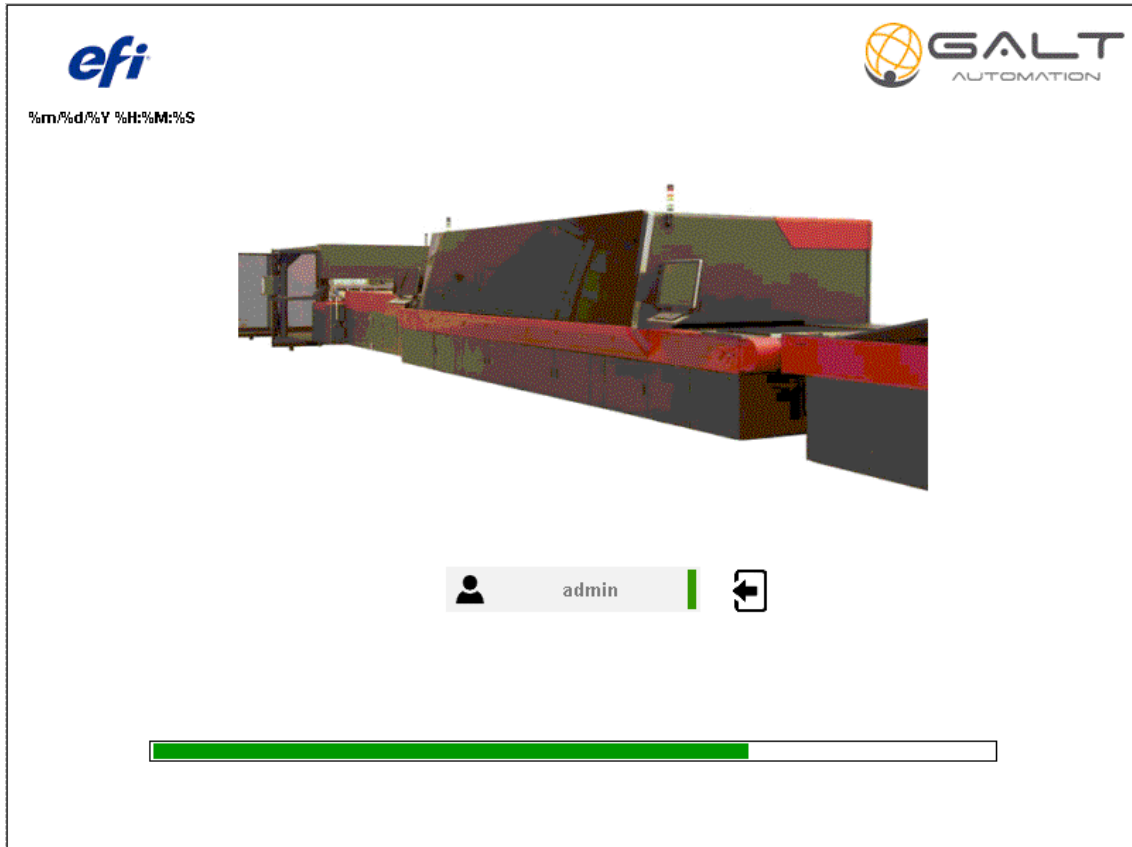
3.1.3. Date and Time Settings

Date and time only can be changed by administrator users from the starting screen. Once the simulation begins, it is no longer possible to change this.



3.2. Information Request

As explained in section 2.1. *User Roles*, there are two ways to operate the simulator: **Printer mode** and **Ancillary mode**. With the exception of the administrators, the rest of the users are limited to a single mode of operation, so in these cases the operating mode is automatically selected when the simulator starts.



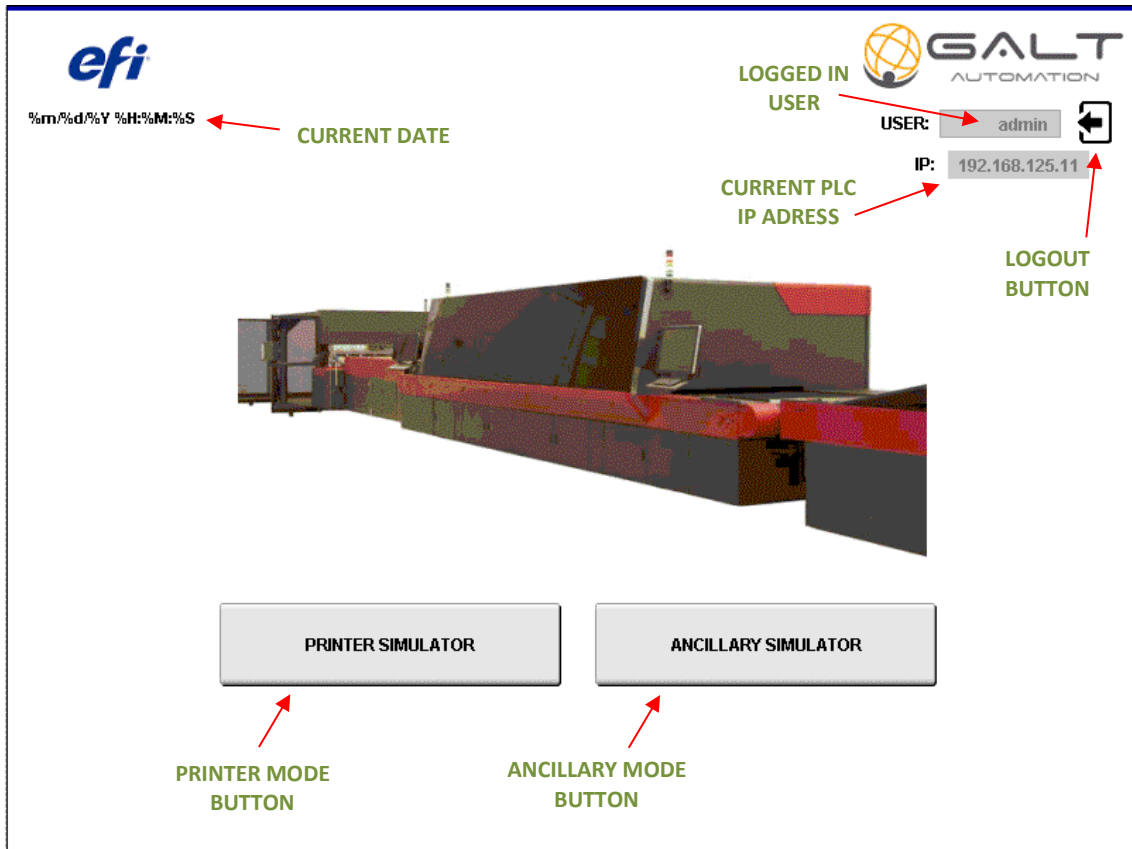
The information request screen will depend on the simulator mode as well as the user role.

- In **printer mode**, it is necessary to enter data about the ancillary machines we want to test; that is, feeder, coater and stacker. Depending on the user role, the user will be able to select between these three machines and enable several of them at the same time, or, on the contrary, the user will be limited to one of them.

- In **ancillary mode**, it is necessary to enter data about the printer machine.

In both cases, it will be necessary to fill in all the enabled fields in order to continue with the simulation. Otherwise, the *continue* button will be locked.

As can be seen in the picture below, administrator users can choose between Printer Mode or Ancillary Mode.



3.2.1. Information Request in Printer Mode

In printer mode, the PLC is set as a printer. In order to communicate with the ancillary machines, it is necessary to specify the IP Address of the ancillary machines we want to test. The PLC IP address can also be set, being important that the PLC does not have the same IP as the ancillary machines.

Depending on the user role, the user will be able to select which machines are enabled during the simulation, or will be limited to one of them. The *continue* button will be locked until the operator fills in all the enabled fields.



%m/%d/%Y %H:%M:%S

PRINTER



USER:

IP:



INFORMATION REQUEST

IP Address: 192.168.125.11		Set
Operator Name: Operator A		
☒ Feeder	Serial Number:	18546FJT12A
	Machine Type:	Feeder
	IP Address:	192.168.125.175
☐ Coater	Serial Number:	-
	Machine Type:	-
	IP Address:	-
☒ Stacker	Serial Number:	14935KCC81F
	Machine Type:	
	IP Address:	192.168.125.185

3.2.2. Information Request in Ancillary Mode

In ancillary mode, the PLC is set as feeder, coater and stacker at the same time; it is possible to enable and disable each of them later. However, Modbus communication cannot be enabled for all ancillary machines simultaneously because, in order to do that, we would need the PLC to have three different IP addresses at the same time.

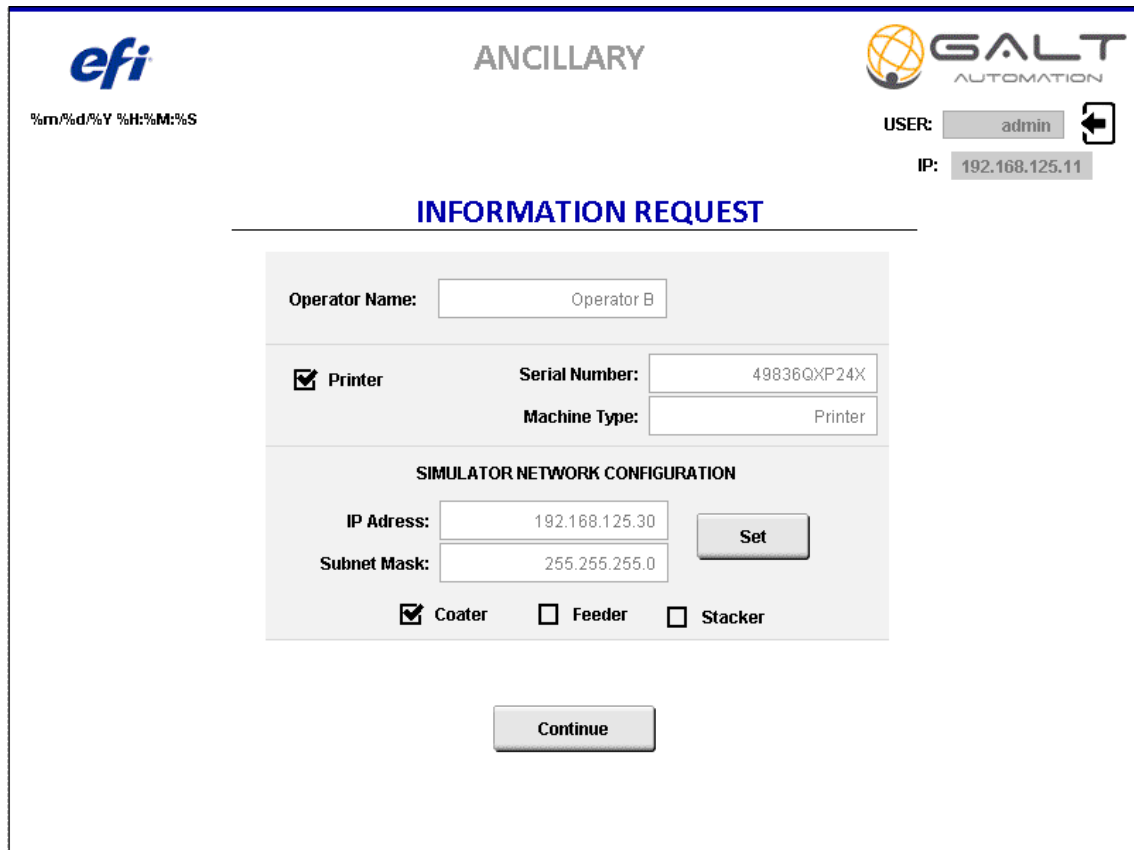
Therefore, the operator has to set the PLC IP address manually according to the IP address that the printer expects for each of the ancillary machines. By default:

- Feeder: 192.168.125.175
- Coater: 192.168.125.30 / 192.168.125.130

- Stacker: 192.168.125.185

Once the PLC IP address has been set, it is important to enable the correct checkbox to indicate to which auxiliary machine the set IP belongs.

As in the previous case, the *continue* button will be locked until the operator fills in all the enabled fields.



The screenshot shows the 'ANCILLARY' configuration interface for GALT Automation. At the top left is the efi logo and a timestamp '%m/%d/%Y %H:%M:%S'. At the top right is the GALT AUTOMATION logo, a 'USER: admin' field with a refresh icon, and an 'IP: 192.168.125.11' field. The main title is 'INFORMATION REQUEST'. The form contains several sections: 'Operator Name' with a text box containing 'Operator B'; a 'Printer' section with a checked checkbox, 'Serial Number' (49836QXP24X), and 'Machine Type' (Printer); a 'SIMULATOR NETWORK CONFIGURATION' section with 'IP Address' (192.168.125.30) and 'Subnet Mask' (255.255.255.0) fields, a 'Set' button, and three checkboxes: 'Coater' (checked), 'Feeder' (unchecked), and 'Stacker' (unchecked). At the bottom is a 'Continue' button.

3.3. Simulation Environment

Regardless of the operating mode, the simulation environment consists of the following screens: Main, Feeder, Coater, Stacker, Results, Encoder and Alarms. However, the Results, Encoder and Alarms screens are not considered in this section and will be explained in later sections.

From the main screen:

- External emergencies are monitored.
- PLC emergencies can be activated.
- *Record*, *enable* and *auto* commands can be switched globally, affecting feeder, coater and stacker screens.
- In printer mode, the buttons that manage the test sequences can also be activated globally.

Feeder, coater and stacker screens collect the information of each of these machines, and are divided into 4 tabs: Progress Diagram, Electrical I/O Status, Modbus Inputs and Modbus Outputs. There is a set of buttons that are common to these tabs: *record*, *enable*, *auto* and *reset*. In printer mode, additional buttons are available to manage the test sequences.

- **Progress Diagram:** Shows, for each of the PLC outputs, the PLC inputs that must be active so that the outputs can be activated. It also monitors the status of these signals.
- **Electrical I/O Status:** Monitors the status of the PLC inputs and outputs. It also allows switching the PLC outputs manually when the AUTO mode is switched off.
- **Modbus Inputs:** Displays the input (from the point of view of the auxiliary machine) registers. Depending on the operating mode, these registers are read only, or read and write.
- **Modbus Outputs:** Displays the output (from the point of view of the auxiliary machine) registers. Depending on the operating mode, these registers are read only, or read and write.

All these screens change their appearance depending on the operation mode.

3.3.1. Simulation Environment in Printer Mode

In printer mode, the PLC simulates the printer, so the emergencies of the feeder, coater and stacker are read and monitored, while the emergency of the printer is simulated and can be activated by a button.

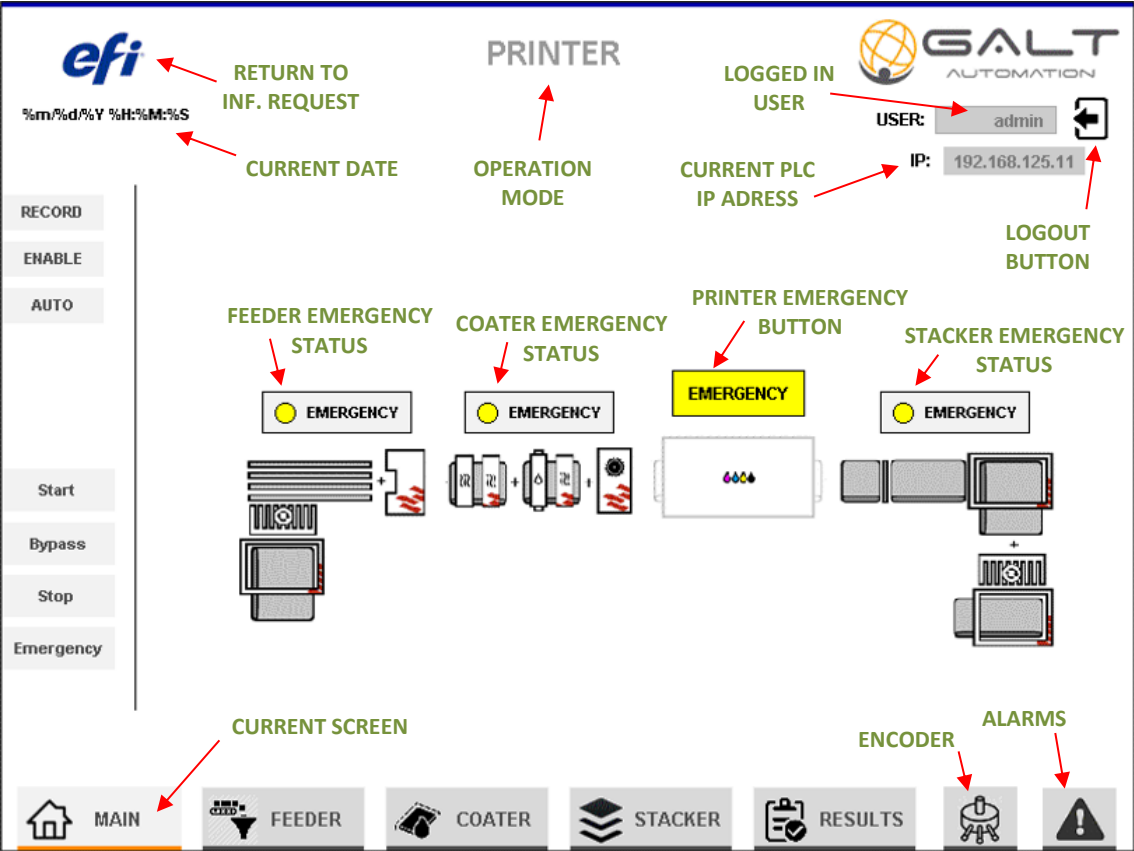
In this case, the goal of the simulation is to check the correct operation of the ancillary machines. Since the testing of these machines can be done together or separately, the *record*, *enable* and *auto* commands have been added for each of them. In printer mode, the function of these commands is as follows:

- **Record:** When enabled, the results of the simulation of that machine are recorded in a CSV file.
- **Enable:** This command enables or disables the simulation of the machine. When enabled, its emergencies are considered and the test sequence is applied to it.
- **Auto:** When enabled, the simulation of this machine is in automatic mode, so it is controlled by the testing sequences. When disabled, PLC outputs are activated manually, regardless of the state of the *enable* command.

When any of these buttons are pressed from the main screen, the change is applied to the three machines. In this way, it is not necessary to enable them one by one, since all of them can be managed simultaneously. By default, these commands are enabled and can only be disabled by users who have an administrator role.

In addition to the above, there is a fourth button that is only managed from the screens of the ancillary machines:

- **Reset:** Reset any sequence error status.

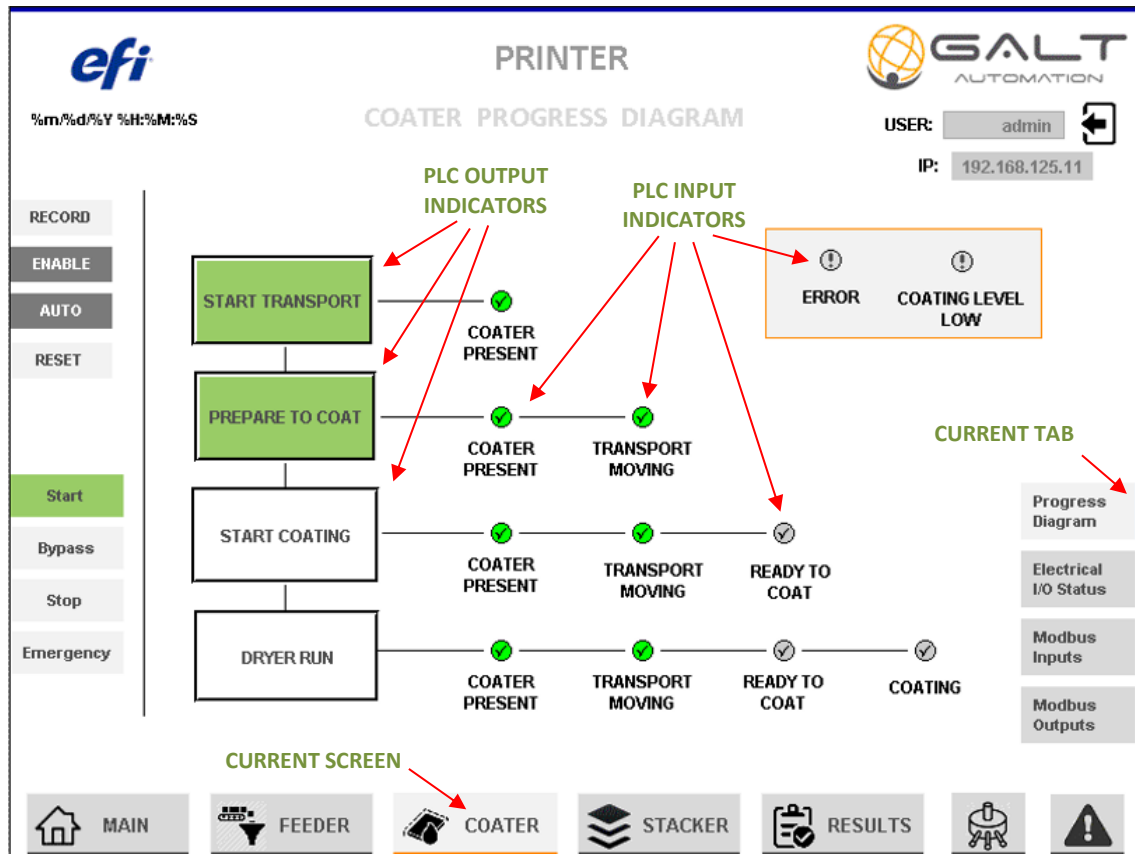


As can be seen above, the main screen has buttons to manage the test sequences. In a similar way to the previous commands, these buttons can be activated globally from the main screen or separately from the respective screens of the ancillary machines.

The sequences only work in automatic mode and with the machine enabled. The following table shows the function of each sequence:

Se quence	Description
Sta rt	Activates the start sequence of the machine

Stop	Activates the stop sequence of the machine
Bypass	Starts the coater without applying the coating
Reject	Activates the reject sequence of the stacker
Stack	Activates the stack sequence of the stacker
Emergency	Activates the emergency stop button of the machine and initiates the emergency stop sequence.



Finally, regarding the modbus tabs, since the PLC is operating in printer mode, it is configured as Modbus client, so it writes to the input registers of the ancillary machines and reads from their output registers. From the Modbus Inputs tab, operators can assign values to each of the registers. From the Modbus Outputs tab, the values of the read registers can be seen.


%m/%d/%Y %H:%M:%S

PRINTER
COATER ELECTRICAL I/O STATUS


USER: admin
IP: 192.168.125.11

RECORD
ENABLE
AUTO
RESET
Start
Bypass
Stop
Emergency

PLC OUTPUTS

☒ START TRANSPORT

☒ PREPARE TO COAT

☐ START COATING

☐ REQUEST DRYER RUN MODE

PLC OUTPUT STATUS

PLC INPUTS

☒ TRANSPORT MOVING

☒ COATER PRESENT

☐ READY TO COAT

☐ COATING

☐ DRYER IN RUN MODE

☐ COATER ERROR

☐ COATING LEVEL LOW

PLC INPUT STATUS

☐ COATER SIMULATION ERROR

Progress Diagram
Electrical I/O Status
Modbus Inputs
Modbus Outputs

MAIN FEEDER COATER STACKER RESULTS


%m/%d/%Y %H:%M:%S

PRINTER
COATER ELECTRICAL I/O STATUS


USER: admin
IP: 192.168.125.11

RECORD
ENABLE
AUTO
RESET

Start
Bypass
Stop
Emergency

PLC OUTPUTS

☒

START TRANSPORT

☒

PREPARE TO COAT

☐

START COATING

☐

REQUEST DRYER RUN MODE

PLC OUTPUT
BUTTONS
(AUTO switched off)

PLC INPUTS

TRANSPORT MOVING ☒

COATER PRESENT ☒

READY TO COAT ☐

COATING ☐

DRYER IN RUN MODE ☐

COATER ERROR ☐

COATING LEVEL LOW ☐

☐ COATER SIMULATION ERROR

Progress
Diagram
Electrical
I/O Status
Modbus
Inputs
Modbus
Outputs

MAIN FEEDER COATER STACKER RESULTS

PRINTER

COATER MODBUS INPUTS

USER: admin

IP: 192.168.125.11

%m/%d/%Y %H:%M:%S

RECORD

ENABLE

AUTO

RESET

Start

Bypass

Stop

Emergency

ADDRESS	DESCRIPTION	VALUE
40000	Media Size - Width [mm*0,1]	0
40001	Media Size - Length [mm*0,1]	0
40002	Media Size - Thickness [mm*0,1]	0
40003	Media - Pitch (Changed to Gap) [mm*0,1]	0
40004	Dryer Run Power Level SP [%]	0
40005	Drying Time (Future) [s]	0
40006	Anilox Run Speed [%]	0
40007	Line Speed SP [MPM]	0
40008	Line Control Word 1	0
40009	Line Control Word 2	0
40010	Line Control Word 3 (Light tower configuration)	0
40011	Dryer Control Word (Optional for Job Recipe)	0
40012	Delay for the dryer to send "dryer run signal" [s]	0
40013	Line Accel Rate SP [MPM/s]	0
40014	Line Decel Rate - Normal SP [MPM/s]	0
40015	Line Decel Rate - Fast SP [MPM/s]	0
40016	Dryer Standby Power Level SP [%]	0
40017	Transfer Conveyor Infeed Blower Speed SP [%]	0
40018	Transfer Conveyor Outfeed Blower Speed SP [%]	0
40019	Sunday Drive Speed SP [MPM]	0
40020	Spare	0
40021	Transfer Roll - Run Position SP [mm]	0
40022	Transfer Roll - Operator Side - Position Offset SP	0
40023	Transfer Roll - Drive Side - Position Offset SP [mm]	0
40024	Coating Tank Low Level SP [%]	0

Progress Diagram

Electrical I/O Status

Modbus Inputs

Modbus Outputs

MAIN

FEEDER

COATER

STACKER

RESULTS



PRINTER

COATER MODBUS OUTPUTS



USER:

IP:

%m/%d/%Y %H:%M:%S

RECORD

ENABLE

AUTO

RESET

Start

Bypass

Stop

Emergency

ADDRESS	DESCRIPTION	VALUE
8192	Line Status Word 1	0
8193	Line Status Word 2	0
8194	Line Status Word 3	0
8195	Dryer Status Word	0
8196	Plate Position Actual [mm]	0
8197	Last Alarm Number Triggered	0
8198	Impression/Transfer Conveyor Drive Status Word	0
8199	Impression/Transfer Conveyor Drive Speed Actual	0
8200	Impression/Transfer Conveyor Drive Torque Actual	0
8201	Anilox/Plate Rolls Drive Status Word	0
8202	Anilox/Plate Rolls Drive Speed Actual [MPM]	0
8203	Anilox/Plate Rolls Drive Torque Actual [%]	0
8204	Transfer Conveyor Infeed Blower Drive Status Word	0
8205	Transfer Conveyor Infeed Blower Drive Speed	0
8206	Transfer Conveyor Infeed Blower Drive Torque	0
8207	Transfer Conveyor Outfeed Blower Drive Status	0
8208	Transfer Conveyor Outfeed Blower Drive Speed	0
8209	Transfer Conveyor Outfeed Blower Drive Torque	0
8210	Alarm Word 01	0
8211	Alarm Word 02	0
8212	Alarm Word 03	0
8213	Alarm Word 04	0
8214	Alarm Word 05	0

Progress Diagram

Electrical I/O Status

Modbus Inputs

Modbus Outputs

MAIN

FEEDER

COATER

STACKER

RESULTS

3.3.2. Simulation Environment in Ancillary Mode

In ancillary mode, the PLC simulates the ancillary machines, so the emergency of the printer is read and monitored, while the emergencies of the ancillary machines are simulated and can be activated by buttons.

In this case, the goal of the simulation is to check the correct operation of the printer. As in the previous case, the testing of the printer can be done separately for each ancillary machine or all together, managed through the *record*, *enable* and *auto* commands. In ancillary mode, the function of these commands is as follows:

- **Record:** When enabled, the results of the simulation of the machine are recorded in a CSV file.

- **Enable:** This command enables or disables the simulation of the machine. When enabled, its emergencies are considered and, if *auto* is enabled, the ancillary machine responds to the signals sent by the printer.
- **Auto:** When enabled, the simulation of this machine is in automatic mode, so the ancillary machine responds automatically to the signals sent by the printer. When disabled, PLC outputs are activated manually, regardless of the state of the *enable* command.

When any of these buttons are pressed from the main screen, the change is applied to all three machines. In this way, it is not necessary to enable them one by one, since all of them can be managed simultaneously. By default, these commands are enabled and can only be disabled by users who have an administrator role.

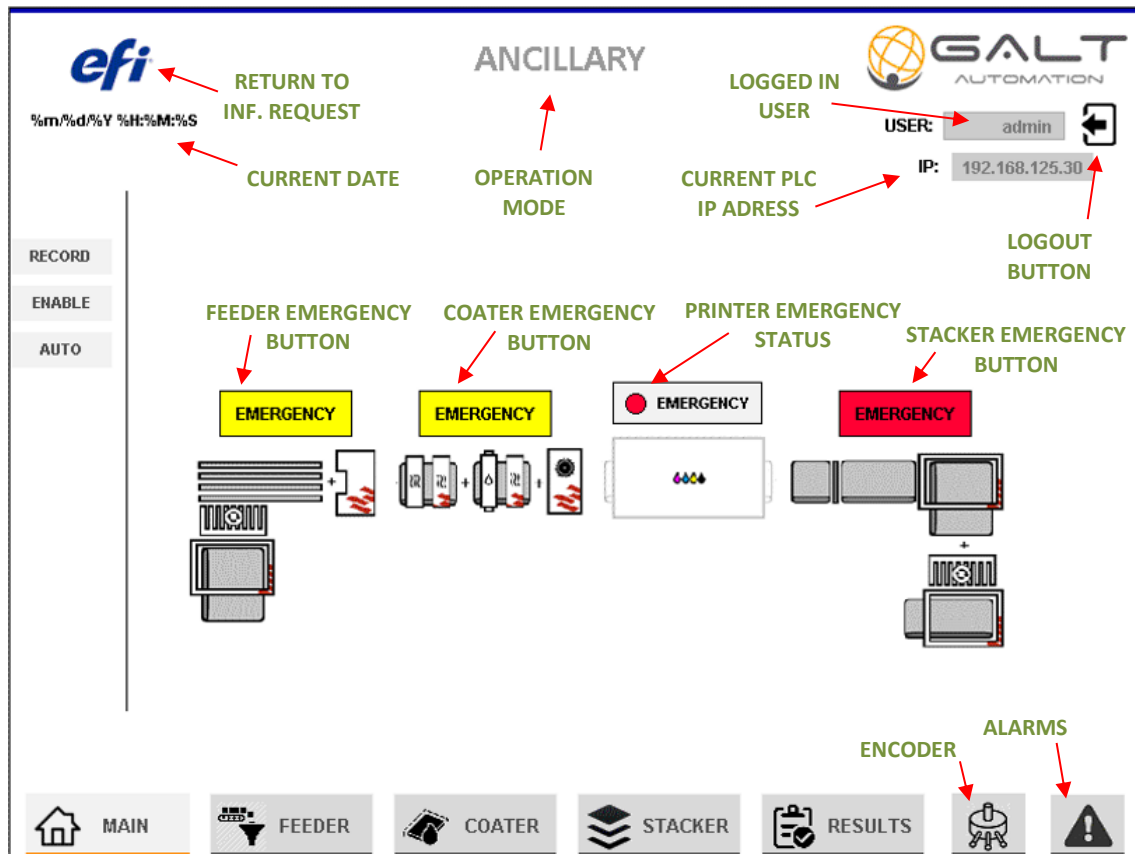
In addition to the above, there is a fourth button that can only be managed from the screens of each ancillary machine:

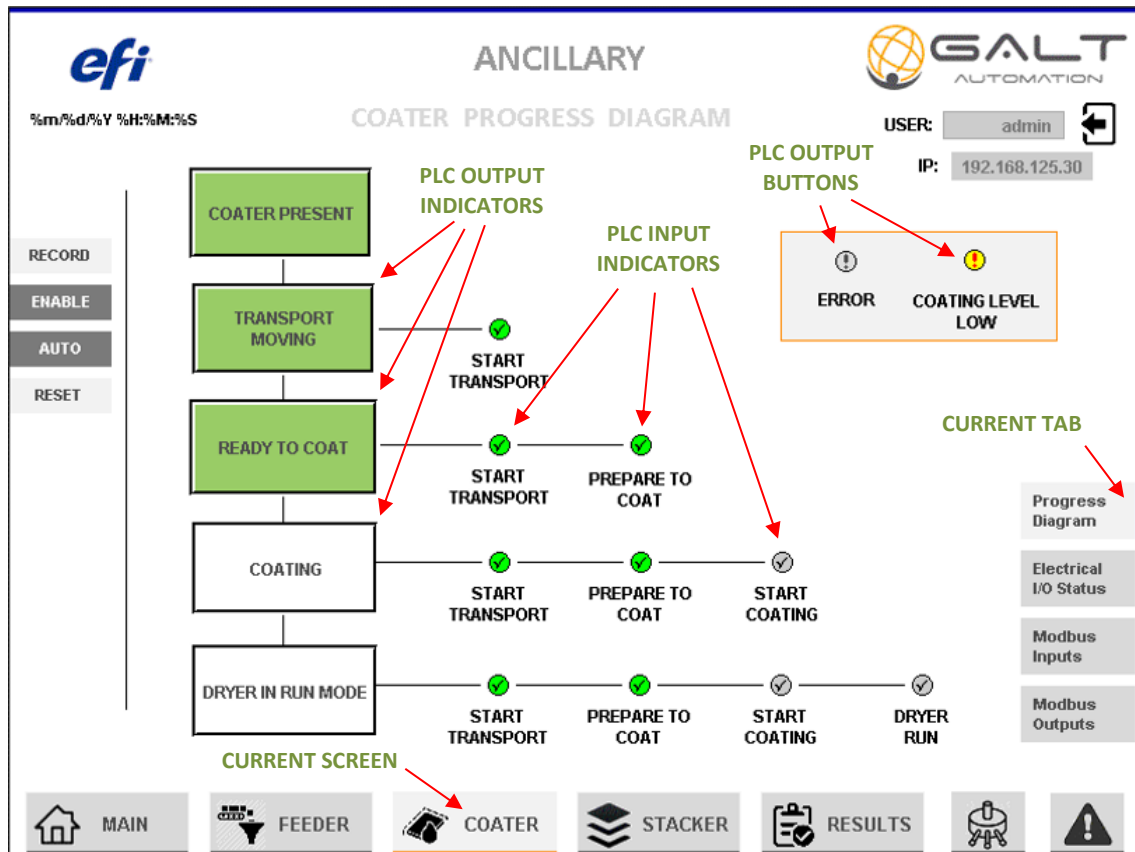
- **Reset:** Reset any sequence error status.

Now, there are no buttons to manage the testing sequences since, in ancillary mode, these sequences are managed from the printer. The PLC switches its outputs according to the signals that the printer sends.

Regarding the modbus tabs, since the PLC is operating in ancillary mode, it is configured as Modbus server, so it writes to the output registers of the ancillary machines and reads from their input registers. From the Modbus Outputs tab, operators can assign values to each of the registers. From the Modbus Inputs tab, the values of the read registers can be seen.

It is important to keep in mind that Modbus communication cannot be enabled for all ancillary machines simultaneously, so the operator has to set the PLC IP address manually to one of the IP address that the printer expects for the ancillary machines. This can be done from the Information Request screen by going back to that screen by clicking on the efi logo.






%m/%d/%Y %H:%M:%S

ANCILLARY
COATER ELECTRICAL I/O STATUS


USER: admin
IP: 192.168.125.30

RECORD
ENABLE
AUTO
RESET

PLC INPUTS

☒ START TRANSPORT

☒ PREPARE TO COAT

☐ START COATING

☐ REQUEST DRYER RUN MODE

PLC INPUT STATUS

PLC OUTPUTS

TRANSPORT MOVING ☒

COATER PRESENT ☒

READY TO COAT ☒

COATING ☐

DRYER IN RUN MODE ☐

COATER ERROR ☐

COATING LEVEL LOW ☐

PLC OUTPUT BUTTONS
(AUTO switched off)

☐ SIMULATION ERROR

Progress Diagram
Electrical I/O Status
Modbus Inputs
Modbus Outputs

MAIN

FEEDER

COATER

STACKER

RESULTS

**ANCILLARY**
COATER MODBUS INPUTS


USER: 
IP:

RECORD

ENABLE

AUTO

RESET

ADDRESS	DESCRIPTION	VALUE
40000	Media Size - Width [mm*0,1]	0
40001	Media Size - Length [mm*0,1]	0
40002	Media Size - Thickness [mm*0,1]	0
40003	Media - Pitch (Changed to Gap) [mm*0,1]	0
40004	Dryer Run Power Level SP [%]	0
40005	Drying Time (Future) [s]	0
40006	Anilox Run Speed [%]	0
40007	Line Speed SP [MPM]	0
40008	Line Control Word 1	0
40009	Line Control Word 2	0
40010	Line Control Word 3 (Light tower configuration)	0
40011	Dryer Control Word (Optional for Job Recipe)	0
40012	Delay for the dryer to send "dryer run signal" [s]	0
40013	Line Accel Rate SP [MPM/s]	0
40014	Line Decel Rate - Normal SP [MPM/s]	0
40015	Line Decel Rate - Fast SP [MPM/s]	0
40016	Dryer Standby Power Level SP [%]	0
40017	Transfer Conveyor Infeed Blower Speed SP [%]	0
40018	Transfer Conveyor Outfeed Blower Speed SP [%]	0
40019	Sunday Drive Speed SP [MPM]	0
40020	Spare	0
40021	Transfer Roll - Run Position SP [mm]	0
40022	Transfer Roll - Operator Side - Position Offset SP	0
40023	Transfer Roll - Drive Side - Position Offset SP [mm]	0
40024	Coating Tank Low Level SP [%]	0

Progress Diagram

Electrical I/O Status

Modbus Inputs

Modbus Outputs

 MAIN

 FEEDER

 COATER

 STACKER

 RESULTS







ANCILLARY

COATER MODBUS OUTPUTS



%m/%d/%Y %H:%M:%S

USER: 

IP:

RECORD

ENABLE

AUTO

RESET

ADDRESS	DESCRIPTION	VALUE
8192	Line Status Word 1	0
8193	Line Status Word 2	0
8194	Line Status Word 3	0
8195	Dryer Status Word	0
8196	Plate Position Actual [mm]	0
8197	Last Alarm Number Triggered	0
8198	Impression/Transfer Conveyor Drive Status Word	0
8199	Impression/Transfer Conveyor Drive Speed Actual	0
8200	Impression/Transfer Conveyor Drive Torque Actual	0
8201	Anilox/Plate Rolls Drive Status Word	0
8202	Anilox/Plate Rolls Drive Speed Actual [MPM]	0
8203	Anilox/Plate Rolls Drive Torque Actual [%]	0
8204	Transfer Conveyor Infeed Blower Drive Status Word	0
8205	Transfer Conveyor Infeed Blower Drive Speed	0
8206	Transfer Conveyor Infeed Blower Drive Torque	0
8207	Transfer Conveyor Outfeed Blower Drive Status	0
8208	Transfer Conveyor Outfeed Blower Drive Speed	0
8209	Transfer Conveyor Outfeed Blower Drive Torque	0
8210	Alarm Word 01	0
8211	Alarm Word 02	0
8212	Alarm Word 03	0
8213	Alarm Word 04	0
8214	Alarm Word 05	0

Progress Diagram

Electrical I/O Status

Modbus Inputs

Modbus Outputs

 MAIN

 FEEDER

 COATER

 STACKER

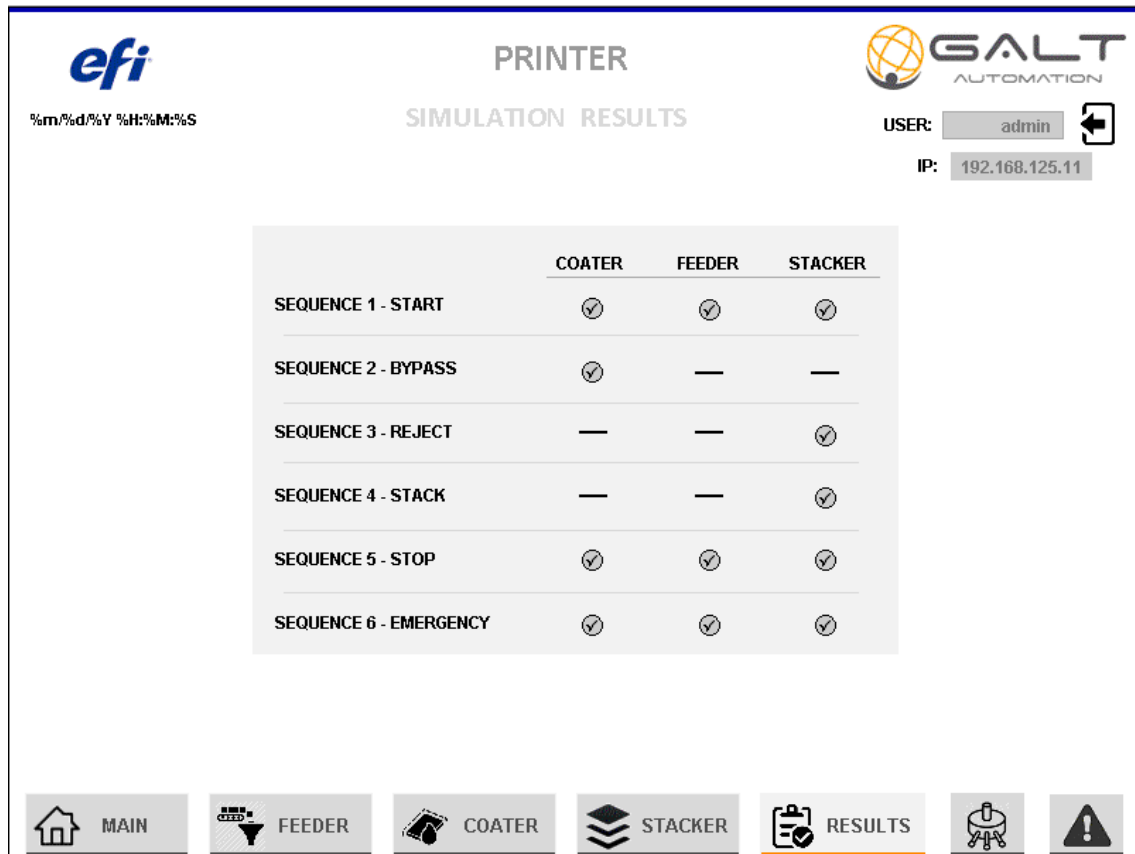
 RESULTS





3.4. Results

The results screen shows the result of the test sequences for each of the ancillary machines.



PRINTER
SIMULATION RESULTS

USER: admin IP: 192.168.125.11

	COATER	FEEDER	STACKER
SEQUENCE 1 - START	✓	✓	✓
SEQUENCE 2 - BYPASS	✓	—	—
SEQUENCE 3 - REJECT	—	—	✓
SEQUENCE 4 - STACK	—	—	✓
SEQUENCE 5 - STOP	✓	✓	✓
SEQUENCE 6 - EMERGENCY	✓	✓	✓

Navigation: MAIN, FEEDER, COATER, STACKER, RESULTS, Warning

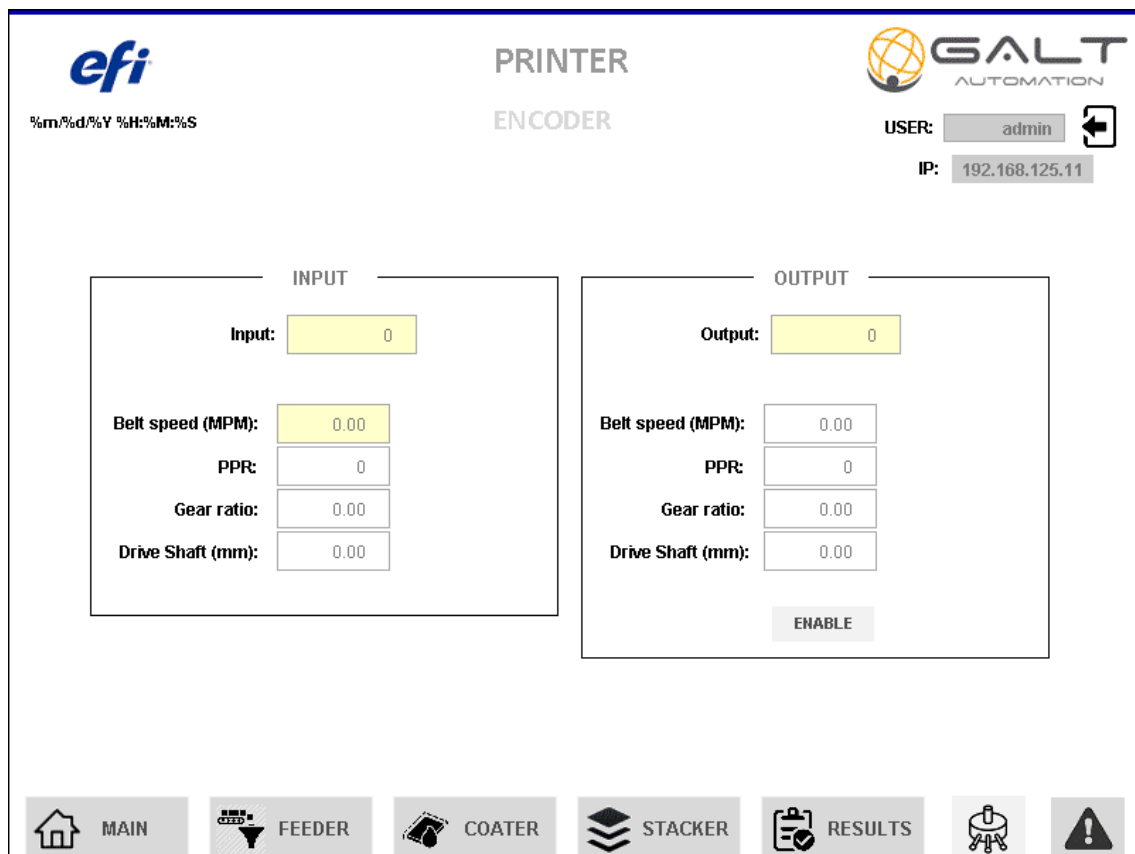
Remember that:

- In printer mode, simulator checks the performance of the feeder, coater and stacker.
- In ancillary mode, simulator checks the correct operation of the printer with each of the ancillary machines.

3.5. Encoder

This screen collects information about the encoders. On the one hand, it shows the reading of the input encoder, coming from the printer. On the other hand, it manages the simulation of the output encoder.

In both cases, the relationship between the encoder pulses and the belt speed depends on the pulses per revolution (PPR) of the encoder, the drive shaft perimeter and the encoder-drive shaft gear ratio.



PRINTER ENCODER

USER: admin IP: 192.168.125.11

INPUT

Input: 0

Belt speed (MPM): 0.00

PPR: 0

Gear ratio: 0.00

Drive Shaft (mm): 0.00

OUTPUT

Output: 0

Belt speed (MPM): 0.00

PPR: 0

Gear ratio: 0.00

Drive Shaft (mm): 0.00

ENABLE

MAIN FEEDER COATER STACKER RESULTS

There are two types of cabinets regarding the encoder card.

Encoder 1 slot

We must change the configuration in the channels card, to allow input encoder or output signal generation. We use the enable button to change it, so we never have both configurations (Input-Output encoder), we need to choose and connect the correct RJ45 wire in the I/O connector.

INPUT	OUTPUT
Input: 0	Output: 0
Belt speed (MPM): 0.00	Belt speed (MPM): 0.00
PPR: 0	PPR: 0
Gear ratio: 0	Gear ratio: 0
Drive Shaft (mm): 0.00	Drive Shaft (mm): 0.00
ENABLE	ENABLE

Encoder 2 slots

We can connect an input encoder signal and an output generation at the same time because we have two cards and two RJ45 connectors. We just need to enable the output generator if we are going to use it. The input signal comes when is connected (we have not enable button).

INPUT	OUTPUT
Input: 0	Output: 0
Belt speed (MPM): 0.00	Belt speed (MPM): 0.00
PPR: 0	PPR: 0
Gear ratio: 0.00	Gear ratio: 0.00
Drive Shaft (mm): 0.00	Drive Shaft (mm): 0.00
	ENABLE

3.6. Alarms

This screen shows the alarm history of the simulator. It is possible to manually remove the alarm indication by acknowledging the alarm by clicking on the acknowledge button.

PRINTER

ALARMS

USER: admin

IP: 192.168.125.11

%m/%d/%Y %H:%M:%S	AlarmText
%m/%d/%Y %H:%M:%S	AlarmText
%m/%d/%Y %H:%M:%S	AlarmText
%m/%d/%Y %H:%M:%S	AlarmText
%m/%d/%Y %H:%M:%S	AlarmText
%m/%d/%Y %H:%M:%S	AlarmText
%m/%d/%Y %H:%M:%S	AlarmText
%m/%d/%Y %H:%M:%S	AlarmText
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%m/%d/%Y %H:%M:%S	AlarmText
%m/%d/%Y %H:%M:%S	AlarmText
%m/%d/%Y %H:%M:%S	AlarmText
%m/%d/%Y %H:%M:%S	AlarmText
%m/%d/%Y %H:%M:%S	AlarmText
%m/%d/%Y %H:%M:%S	AlarmText
%m/%d/%Y %H:%M:%S	AlarmText

ACKNOWLEDGE
BUTTON

MAIN

FEEDER

COATER

STACKER

RESULTS